A rival that grew the market: twenty years of KTX in open data

Korean high-speed rail carried 20 million passengers in 2004, its opening year (from April), and 117 million in 2024. On the way it gained a rival operator, new lines and a pandemic. An interactive page and six Python notebooks follow KORAIL's open data from 2004 to 2025. Did the rival split the market or grow it, did new lines create travel or move it, and has everything recovered from COVID?



The station explorer for 부산 (Busan). From 2017 the orange SRT bars sit on top of a KTX bar that has shrunk, and the two together are taller than KTX alone ever was.

Lik Ren Tai, November 2026

Open the interactive page (also on phones)
(https://likrentai.github.io/koreanrail/)

Code, notebooks and data notes on GitHub
(https://github.com/likrentai/koreanrail)

The question

The KTX opened in April 2004 on the Seoul to Busan and Seoul to Mokpo corridors. Since then the network has added the Honam high-speed line (2015), the Gangneung line built for the PyeongChang Winter Olympics (2017) and, in December 2016, a second operator: SR runs the SRT from Suseo in south-east Seoul on the same high-speed tracks. Then COVID emptied the trains for two years.

Each of these is a question that transport economists and planners ask everywhere. When a **rival** enters, does it take passengers from the incumbent or bring new ones? When a **new line** opens, does it create travel or only move riders from slower trains? And after a shock, does demand **return to its old path**? I wanted to answer them with KORAIL's own open data, and to compare like with like: the same stations before and after, and a comparison group that the change did not touch.

The data

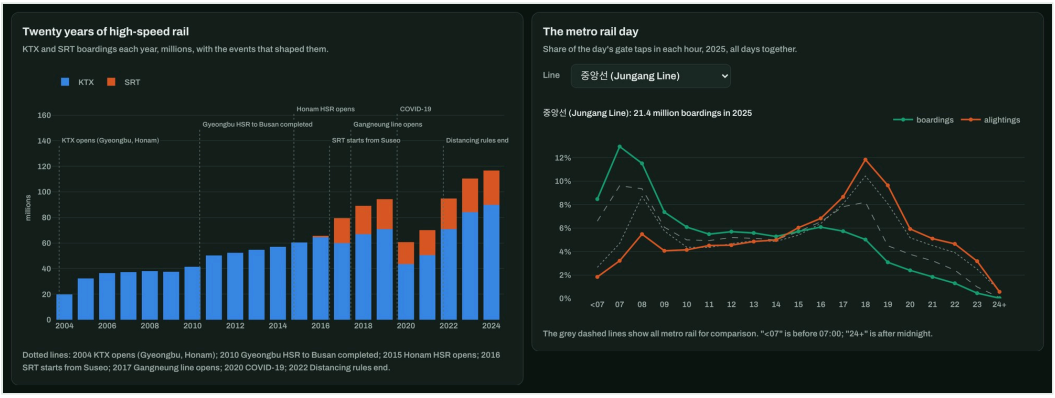
What	Source	Period
KTX boardings plus alightings by station, monthly (50 stations)	KORAIL, Public Data Portal (data.go.kr)	Apr 2004 to Dec 2022
KTX and SRT boardings and alightings by station and direction, yearly	Korean Railway Statistics (railstat.korail.com)	2004 to 2024
Riders by train type, monthly; riders by line and train type, yearly	Korean Railway Statistics	2016 to 2023; 2004 to 2024

Scheduled train runs and train km	Korean Railway Statistics	2016 to 2024
Railway accidents and service disruptions, monthly	Korean Railway Statistics	2016 to 2024
Metro rail (광역철도) boardings by station, monthly, and gate taps by hour	KORAIL metro rail statistics (info.korail.com)	2019, 2024, 2025

Sixteen files in all, cleaned and checked in six notebooks: station totals add up to each table's own subtotals, monthly rows to yearly rows, and KORAIL's yearly totals match its published figures exactly (136.7 million riders in 2023, 144.8 million in 2024). Two numbers are not measurements: the **pre-COVID path** is a model fitted to 2017 to 2019, and the unit of the **train km** table (per day) is inferred, because the export does not state it. Station names are given in Hangul with English in brackets.

One page, six notebooks

Part	What it answers
Stations tab	How many KTX and SRT riders use a station, how that changed each year and month, and what share SRT has. All 88 stations, Hangul with English names.
Timeline tab	Twenty years of high-speed riders with the events that shaped them, and the month-by-month story of SRT's arrival and COVID.
Metro day tab	When metro rail riders travel, line by line, and which lines are most commuter-heavy.
Notebooks (Python)	00 to 02 load and check the sixteen files; 03 explores five candidate findings; 04 makes the charts below; 05 writes the data for the page.



Left: the Timeline tab, with the events that shaped high-speed rail. Right: the Metro day tab for 중앙선 (Jungang Line), the most commuter-heavy line, against all metro rail.

What the data shows

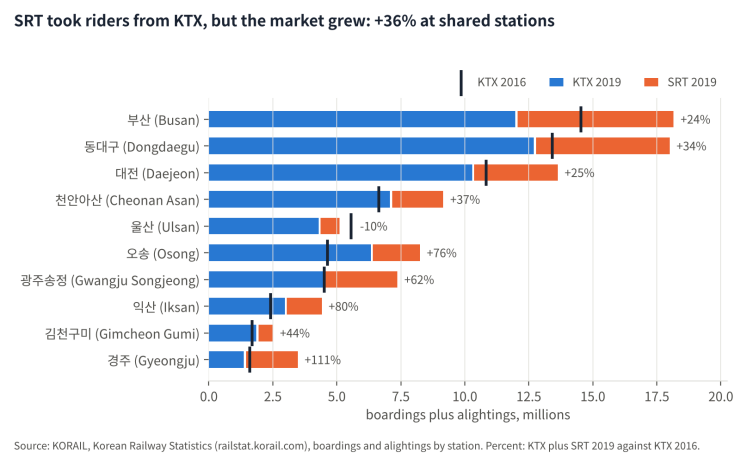
SRT took riders from KTX, but the market grew

+36%

KTX plus SRT riders at the 14 stations SRT served in 2019, against KTX alone in 2016 (KTX by itself: -3%)

At the stations the two operators share, KTX lost riders after SRT arrived: 부산 (Busan) by 18% between 2016 and 2019, 동대구 (Dongdaegu) by 5%. But the two together carried far more than KTX had alone: 24% more at Busan, 34% more at Dongdaegu, 36% more across all 14 shared stations. KTX stations that SRT does not serve grew by 12% over the same years, so the growth at shared stations was three times the background. Nationally, KTX fell from 64.6 to 59.8 million riders in 2017, SRT's first full year, while high-speed rail as a whole rose from 65.6 to 79.3 million.

The data cannot say why, but a likely part of the answer is access: Suseo put a high-speed terminal in south-east Seoul, far from Seoul station and Yongsan. 울산 (Ulsan) is the exception, 10% down.



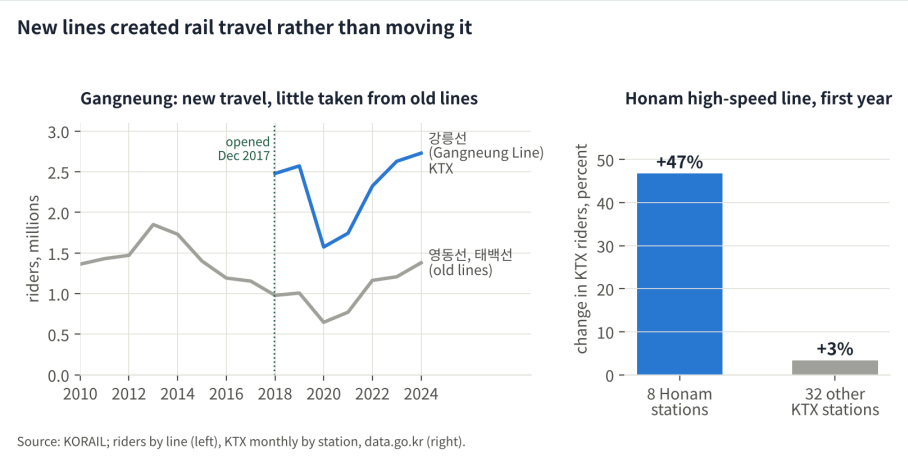
Riders in 2019 at the ten busiest shared stations, KTX and SRT, against KTX's own level in 2016 (the dark tick). Percentages compare the whole bar with the tick.

New lines created rail travel rather than moving it

2.6 million

riders on the Gangneung line in 2019, while the old lines to the east coast lost 0.2 million

Before the Gangneung KTX opened in December 2017, rail to Gangneung meant slow Mugunghwa and Saemaul trains on the 영동선 (Yeongdong) and 태백선 (Taebaek) lines. If the KTX had only moved those riders, the old lines would have lost about as many as the new line gained. They lost 0.18 million between 2016 and 2019 (1.19 to 1.01 million), against 2.57 million on the new line: almost all of it was new to rail, whether from roads or new trips. The Honam high-speed line tells the same story. In the twelve months after it opened on 2 April 2015, KTX riders at the eight Honam stations rose by 47%, against 3% at the other 32 KTX stations over the same months.



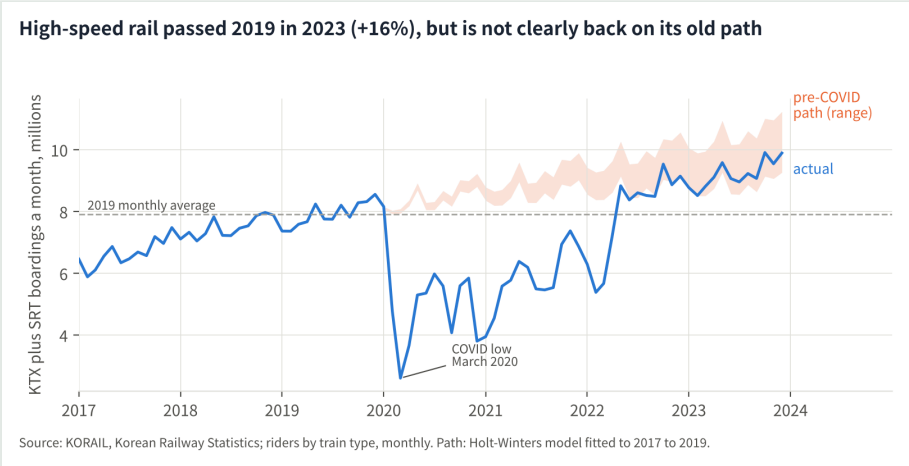
Left: the Gangneung line and the old lines it could have drawn from. Right: KTX riders in the first year of the Honam high-speed line against the year before, at the Honam stations and at all other KTX stations.

High-speed rail is above 2019; metro rail is not

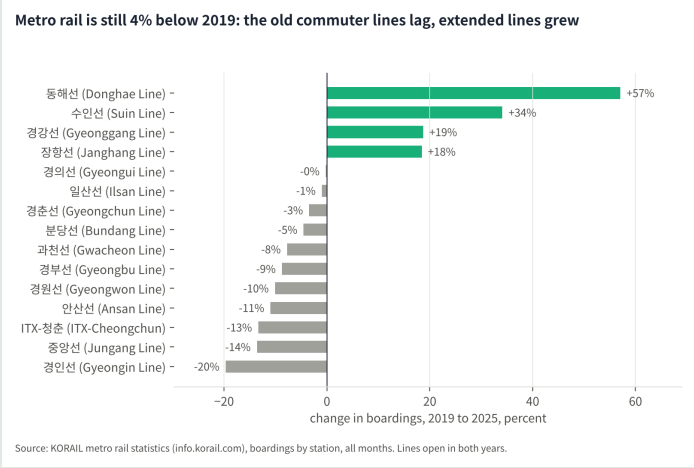
+24% and -4%

high-speed riders in 2024 against 2019 (116.6 against 94.2 million), and metro rail riders in 2025 against 2019

COVID cut high-speed riders by more than a third in 2020, and by about two thirds in the worst month. Against the path the market was on before 2020, an estimated 81 to 105 million trips were lost between 2020 and 2022. Riders passed the 2019 level in 2023 (+16%), but whether they are back on the old path depends on how the trend is drawn: a straight trend puts 2024 13% below it, a levelling-off trend 9% above it, so the chart shows a range rather than one line. Metro rail is a different story: 782.8 million boardings in 2025, 3.9% below 2019. The old commuter lines lag (경인선 Gyeongin -20%, 중앙선 Jungang -14%, 경부선 Gyeongbu -9%), and all the growth is on lines extended or linked since 2019 (동해선 Donghae +57%, 수인선 Suin +34%). Long-distance travel has come back faster than the daily commute.



KTX plus SRT riders each month against the pre-COVID path, drawn as the range between a straight and a levelling-off trend.



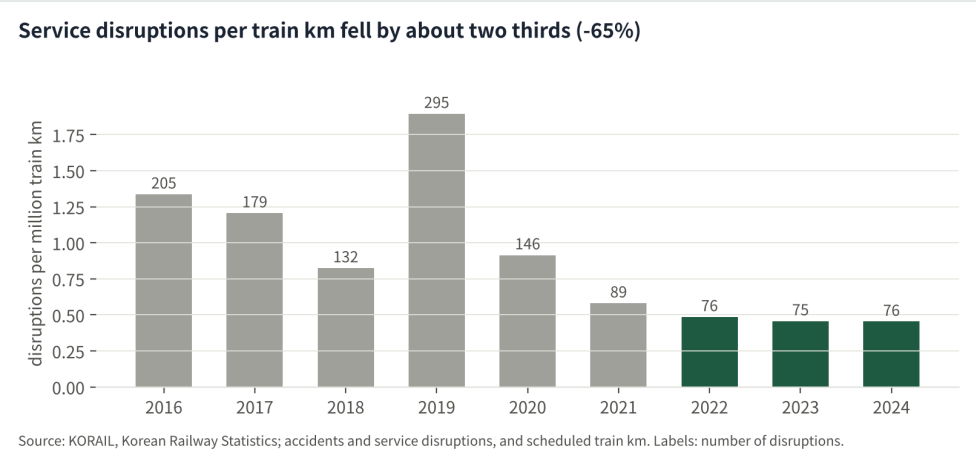
Metro rail boardings by line, 2019 to 2025, for lines open in both years.

Service disruptions per train km fell by about two thirds

-65% service disruptions per million train km, 2022 to 2024 against 2016 to 2019

No punctuality figures are published, so the closest open measure of reliability is KORAIL's count of service disruptions (운행장애): events that stop or delay trains without an accident, such as vehicle, signal or power faults. They peaked at 295 in

2019, 169 of them vehicle faults, and have stayed at 75 to 76 a year since 2022, while train km rose. Per million train km the rate fell from 1.90 in 2019 to 0.45 in 2024. The data does not say why, and railway accidents (collisions, derailments, level crossings) show no such fall: between 9 and 28 a year.



Service disruptions per million train km; the labels give the number of disruptions in each year.

Problems in the data, and how they were handled

- **Two station series that do not quite meet.** The monthly file covers 50 KTX stations to 2022, shows stations under 500 riders a month as 0, and rounds to thousands; the yearly portal table covers every station to 2024. Where they overlap, stations agree within 1%, with two exceptions: Seoul (below) and 행신 (Haengsin) in 2022 (1.4%). The monthly file is used to 2022 and the portal table after it.
- **Seoul counted two ways.** From 2018 the portal's figure for 서울 (Seoul) is 7% to 10% higher than the monthly file, because it includes Gangneung line KTX riders and the monthly file does not. Both are shown, with a note.
- **Renamed stations.** 신경주 became 경주 (Gyeongju), 신포항 became 포항 (Pohang), and the airport station changed name twice. Old names are mapped to today's names before any comparison.
- **An incomplete year.** The portal's 2004 to 2015 station table is incomplete for 2015 (Busan 8.5 against 14.5 million), probably because line codes changed when the Honam line opened. It is used only as a check, for 2004 to 2014.
- **Small mismatches between tables.** SRT in the monthly table is 1% to 3% higher than in the station table for 2017 to 2019; the 2016 network total differs by 2.1 million between two tables. Each comparison uses one table throughout.
- **Undocumented units and labels.** The train runs table does not state its unit; it behaves like trains and train km per day. The hourly metro sheets do not say whether "08시" means the hour starting or ending at 8; the start of the hour fits the data. One sheet is titled "1900년 1월" by mistake; months are taken from the sheet tabs.
- **Messy exports.** Some portal CSVs end without a closing quote, workbooks are laid out for printing, and similar-looking tables are easy to mix up (the first download was the all-trains table, not KTX). Every file is checked against its own subtotals.

Limits

These are before-and-after comparisons with comparison groups, not controlled experiments: other things changed at the same time, such as fares, timetables, roads and the economy, and the data has no origin-to-destination trips, so it cannot show

where new riders came from. The pre-COVID path rests on three years of data and is shown as a range for that reason. Station counts are boardings plus alightings, not people. The metro hours mix weekdays and weekends. Revenue, fares and the operators' costs are outside this project, and nothing here has been through peer review.

Where this could be useful

- **Rail operators and regulators:** evidence on what a second operator did to the market, station by station, rather than only nationally.
- **Transport planners:** a simple test of whether a new line creates travel, with a comparison group, for business cases and reviews after opening.
- **Analytics and consulting teams:** a worked example of turning messy official exports into checked tables, findings with honest ranges, and a page that non-specialists can explore.

The same approach transfers to any network that publishes station counts, such as the Klang Valley rail data in the first project of this series, and to new lines anywhere, by comparing their stations with the rest of the network before and after opening.

About

Lik Ren Tai applies data science in two fields: environmental science and railway systems. This project was built with Python (pandas, openpyxl, statsmodels, matplotlib) and Plotly.

Data: Korea Railroad Corporation (KORAIL), through the Public Data Portal (data.go.kr), the Korean Railway Statistics portal (railstat.korail.com) and KORAIL's metro rail statistics (info.korail.com); SRT figures as published in the Korean Railway Statistics. Fonts: Archivo, Source Serif 4 and Noto Sans KR, SIL Open Font License. This is an independent project, not affiliated with or endorsed by KORAIL, SR or any of the sources.